

SUSPENSE SUSPENSE
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Are You In Suspense?

Are you wondering what happened to your order or payment, even after a fortnight? But did you mention your name and address clearly while sending payment, and what the payment was for?

We get several payments every month without adequate instructions or even identity of the person sending the payment. As a result we are unable to take appropriate action on these, till we get a complaint - if at all.

At EFY, we want you to be our satisfied customer. So please do let us know in case your payment or instructions are not even acknowledged within a fortnight. We are not that lazy!

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an advanced monitoring system that could be mounted directly on an oil well servicing pump in a rugged environment, to perform advanced analysis on sensor data," adds Raghavan.

Power equipment. Power equipment such as UPS are also available that can operate at extreme temperatures, when deployed in military outposts or industrial environments.

Sriram Ramakrishnan, CEO of Consul Consolidated, an Indian player in this space, says, "Consul offers UPS systems and inverters that can withstand extreme temperatures. These are custom designed and manufactured to meet the requirements of the order. Consul has supplied custom-built UPS for various defence applications to operate with defence equipment in extreme temperature conditions, as per JSS specifications."

All components used in defence applications meet MIL-grade specifications instead of commercial grade to operate in extreme temperature conditions. All the cables used in the cabinet are of shielded type for reducing the EMI and RFI interference.

Due to the need to operate in extreme conditions and other special needs, the cabinets are also constructed of higher-grade steel, with special surface treatment for conductivity. The layout of the internal components has also been changed to meet the RFI and EMI specifications.

"Apart from materials, you also need to consider self-contained cooling and ventilation for the UPS system to operate successfully," adds Ramakrishnan.

Consul has also worked closely with DRDO on some of DRDO's missile projects, and with the Brahmos team to create prototypes of UPS systems to be used in the early stages of the project. They have worked right from the design stage to the fabrication of prototypes and field testing of the UPS.

More to come

There are many more such extreme-temperature products available in the

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market today. There is also a lot of research happening across the world to understand and improve low- and high-temperature operation of electronics, hermetic packaging, cryogenic electronics, etc.

For example, in March 2013, Sandia National Laboratories (SNL) announced that it has developed a gold-silver-germanium alloy that is capable of withstanding high temperatures and pressures. The melting point of the alloy is around 450°C, making it suitable for use as a solder for electronics in high-temperature environments, such as those in geothermal wells. Current brazing processes occur at temperatures above 750°C, and the materials used are generally not suitable for electronics. Similarly, soldering occurs at temperatures below 350°C, which is also unsuitable for high-temperature electronics. Therefore SNL believes that its new alloy will fill a niche for high-temperature soldering.

Similarly, last year scientists at the University of Illinois, Urbana-Champaign, discovered the self-cooling effects of graphene transistors, which could do away with the need for heat sinks and other cooling mechanisms in high-temperature operating environments.

As electronics finds its way into more and more niche applications, there will be all the more need for it to work in cooler and hotter environments—and more scope to put specialised extreme-temperature electronics and research prototypes to good use! ●

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